EMPLOYEE HIERARCHY MANAGEMENT SYSTEM

Using Tree Data Structure in C

Mahesh Kumar Reddy M

Goutham P

B TECH – COMPUTER SCIENCE & ENGINEERING

SAI UNIVERSITY

PROBLEM STATEMENT

- Organizations follow a hierarchical employee structure
- Managing employee relationships manually is difficult
- Need an efficient system to store and manage hierarchy
- Perform Create, Read, Update, and Delete operations

REAL-WORLD APPLICATION

- Corporate organization charts
- HR management systems
- University staff hierarchy
- Government administrative structures

DATA STRUCTURE USED

- General Tree
- Implemented using Child–Sibling Representation
- Each employee can have multiple subordinates
- Efficient and flexible hierarchy management

SYSTEM DESIGN

Each Employee Node Contains:

- Employee ID
- Name
- Designation
- Pointer to first child (subordinate)
- Pointer to next sibling

CRUD OPERATIONS IMPLEMENTED

- Create – Add a new employee
- Read – Display employee hierarchy
- Update – Modify employee details
- Delete – Remove employee and subordinates
- Search – Find employee by ID

ALGORITHM EXPLANATION

- Depth First Search (DFS) traversal
- Recursive approach
- Menu-driven execution
- Dynamic memory allocation

IMPLEMENTATION DETAILS

- Programming Language: C
- Concepts Used:

  - Structures
  - Pointers
  - Dynamic Memory (malloc, free)
  - Recursion
  - Modular programming

SAMPLE OUTPUT

```
ID:1 | Mahesh   | CEO
├── ID:2 | Gowtham   | Manager
│   ├── ID:4 | Venky | Developer
│   └── ID:5 | Varun    | Tester
└── ID:3 | Santhosh | HR
```

CONCLUSION

- Successfully implemented employee hierarchy system
- Tree data structure used effectively
- All CRUD operations implemented
- Project demonstrates real-world application of data structures

THANK YOU